

Stability-Aware Alignment for Single-Source Domain Generalization in Medical Image Segmentation

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Abstract

Single-source domain generalization (SSDG) for medical image segmentation aims to learn a robust model from a single annotated source domain that generalizes well to unseen target domains. Recent advances mainly rely on augmentation and invariant representation learning. However, existing methods often impose uniform consistency constraints and implicitly assume that all perturbed regions are equally reliable for alignment. Such uniform alignment may amplify unstable or noisy regions, especially under strong style perturbations, thereby limiting generalization. In this paper, we propose a Stability-Aware Alignment (SAA) method that explicitly models cross-view semantic stability for feature alignment. Specifically, we first design a tri-view perturbation scheme to induce diverse shifts and estimate region-wise stability from cross-view consistency. We then introduce a stability-guided region selection mechanism to identify reliable anatomical regions, followed by a distance-weighted alignment strategy that modulates alignment strength according to stability confidence. Furthermore, we incorporate a structure–style decoupling module to suppress domain-specific appearance variations while preserving task-relevant structural representations. By coupling stability estimation with alignment regulation, our method avoids over-constraining unstable regions and promotes learning of domain-invariant structural features. Extensive experiments on cross-modality and cross-sequence medical image segmentation benchmarks demonstrate that SAA consistently outperforms state-of-the-art SSDG methods, yielding superior robustness on unseen domains.

CCS Concepts

• Computing methodologies → Image segmentation; Machine learning.

Keywords

Single-source domain generalization, Medical image segmentation, Stability-aware alignment, Robust representation learning

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1 Introduction

Medical image segmentation is fundamental to many clinical applications, including diagnosis, treatment planning, and outcome evaluation. Recently, deep-learning-based methods have achieved remarkable success in this task [12, 33]. However, most methods rely on the assumption that the training and test data are drawn from the same distribution. In practice, medical images acquired across institutions, scanners, and protocols often exhibit substantial appearance variations, causing domain shifts that significantly degrade segmentation performance on unseen targets [14, 27, 37].

To mitigate domain shifts, prior studies have explored domain adaptation and multi-source domain generalization [9, 46]. However, access to unlabeled target-domain data or multiple source domains is often impractical in medical scenarios due to privacy, regulatory, and annotation constraints. This makes single-source domain generalization (SSDG), which learns from only one source domain and generalizes to unseen targets, a more realistic yet more challenging setting. The core challenge of SSDG is to bridge the gap between source and unseen target domains. In medical imaging, anatomical structure is generally consistent, whereas intensity, contrast, texture, and noise may vary substantially across domains. These variations are largely style-related rather than label-governing semantics [30, 32]. Therefore, SSDG for medical image segmentation should encourage the model to rely on structure-relevant, domain-stable cues while reducing sensitivity to appearance variations and spurious context correlations [30]. This requires not only increasing appearance diversity during training, but also identifying which semantic responses remain reliable when perturbations become stronger.

Existing SSDG methods for medical image segmentation mainly follow two paradigms: input-level augmentation and representation-level invariant learning [3, 15]. The former increases appearance diversity through style perturbation, saliency-guided augmentation, or frequency-domain manipulation [18, 30, 32, 35], but remains constrained by source-defined transformations and cannot fully cover unseen target shifts. The latter improves robustness by enforcing consistency across perturbed views or imposing invariant regularization in feature space [28, 41, 42]. However, both paradigms largely overlook a critical question: which semantic responses remain stable under perturbation, and which do not? In practice, perturbations often induce unstable boundary responses and fluctuating contextual activations. As a result, indiscriminate global alignment may mix reliable anatomical cues with unstable signals and propagate noise into representation learning, as illustrated in Fig. 1. Moreover, different spatial regions do not contribute equally to robust generalization: anatomically meaningful regions are typically more task-relevant, whereas background or non-interest regions may contain domain-specific cues that encourage shortcut learning [21, 45, 48]. Recent attempts to reweight alignment by uncertainty or foreground priors partially alleviate this issue [38, 45].

However, they still do not explicitly measure whether a region remains semantically stable across diverse perturbations, nor do they indicate which regions should be preferentially aligned when perturbation strength varies.

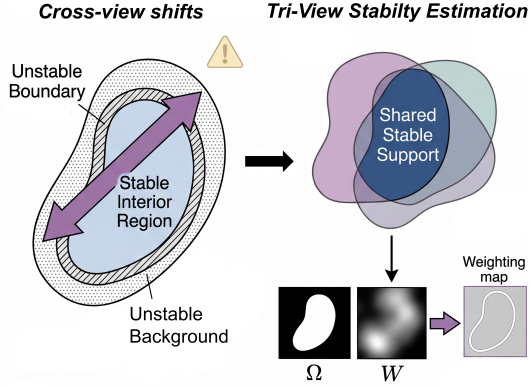


Figure 1: Illustration of cross-view semantic instability in consistency learning. Under diverse perturbations, some spatial regions, particularly boundaries and background, become less consistent across views. A tri-view consensus extracts shared stable regions and converts them into a weighting map for selective alignment.

In this paper, we define semantic stability as the repeatability of representations or predictions for a semantic region under diverse perturbations. Based on this view, we propose Stability-Aware Alignment (SAA), which explicitly models cross-view semantic stability to guide selective alignment. Specifically, SAA is built upon tri-view perturbation, channel-wise structure-style decoupling, and stability-aware selective alignment. These designs estimate stability across complementary perturbed views, select reliable regions for alignment, and adaptively modulate alignment strength according to semantic consistency. Meanwhile, a channel-wise structure-style decoupling module further suppresses appearance-sensitive channels and improves the reliability of stability modeling. Extensive experiments on cross-modality and cross-sequence benchmarks demonstrate that SAA consistently outperforms state-of-the-art SSDG methods. The main contributions of this work are summarized as follows:

- We propose SAA, a stability-aware framework for SSDG in medical image segmentation that explicitly models cross-view semantic stability.
- We design a selective alignment mechanism that estimates semantic stability to determine where and how strongly alignment should be enforced.
- We introduce a channel-wise structure–style decoupling module to suppress appearance-sensitive channels and improve stability-aware representation learning.
- Extensive experiments on cross-modality and cross-sequence benchmarks demonstrate the effectiveness of the proposed method.

2 Related Works

2.1 Single-Source Domain Generalization for Medical Image Segmentation

Domain generalization for medical image segmentation aims to ensure robust performance under acquisition-induced shifts without access to target-domain data, and is often further constrained to the single-source regime due to privacy and annotation costs [46]. Recent SSDG approaches mainly follow two paradigms: (i) augmentation-driven domain expansion and (ii) domain-invariant feature learning [15, 28]. The first paradigm expands the source distribution via diverse appearance variations. Representative strategies include saliency-guided perturbations [35], frequency-based manipulations [19, 32, 49], input-space interpolation [44], and causal or shortcut-breaking perturbations [30]. Other Fourier-based style diversification modules, such as MoreStyle [49], further relax frequency constraints to generate richer appearance variations. While effective at increasing input diversity, these approaches remain limited to source-defined transformations and may distort structural semantics [23, 35, 42]. The second paradigm promotes invariance in the prediction or feature space by reducing sensitivity to domain-specific appearance factors. Representative examples include cross-view consistency in IRLSG [28], channel- or style-disentanglement methods, as in C2SDG and ICRN[5, 10], multi-receptive-field feature disentanglement in MRFFD [41], and invariant-representation learning in EIR-SDG through texture–style diversification, adaptive whitening, and semantic relationship optimization [29]. More recent studies further combine augmentation with explicit representation constraints, as seen in FSDA-DG and DCON [38, 45]. Overall, recent robust segmentation research increasingly combines multi-view perturbations with representation alignment [26, 34], reflecting a shift toward learning stable, domain-invariant representations rather than relying solely on input diversification [8, 24].

2.2 Multi-view Consistency and Reliability Aware Supervision

A key challenge in invariance learning is that consistency is only as reliable as its supervisory signal. Many methods enforce agreement across perturbed views in prediction or feature space [48], yet noisy predictions or perturbed responses can be amplified under strong perturbations or domain shifts [1, 2, 10]. Such noise amplification effects are well-documented in teacher–student and pseudo-labeling frameworks, where indiscriminate consistency can lead to confirmation bias [20, 36, 40]. Existing efforts mainly address this issue in three ways. Some improve target quality directly, for example by constructing denoised semantic supervision from synthetic semantic stacks [31]. Others modulate consistency using reliability proxies such as uncertainty, entropy, or confidence [7, 25, 47]; representative examples include uncertainty-weighted consistency in FSDA-DG and related uncertainty-aware weighting schemes [39, 45]. A third line of work restricts consistency to informative spatial regions, such as object-centric foreground areas in DCON [38] or ambiguous and boundary-related regions in MPDC [51], rather than the full image [22].

Despite these advances, two limitations remain. First, existing methods mainly rely on prediction-based reliability estimates rather

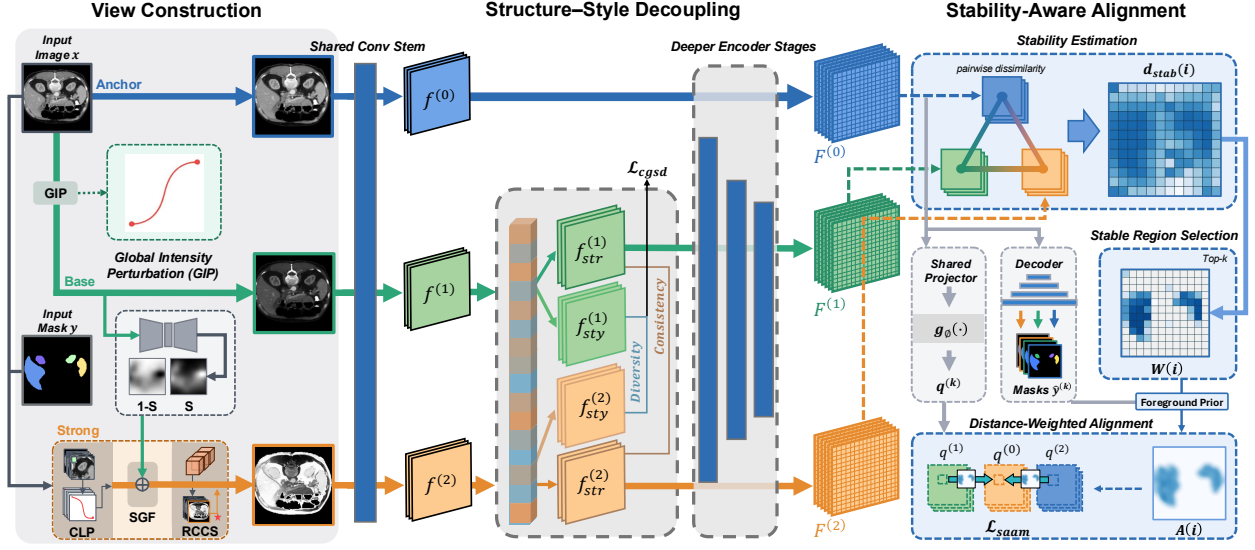


Figure 2: Overview of the proposed Stability-Aware Alignment (SAA) method. SAA includes three modules: (1) Tri-View Construction for complementary inputs, (2) CGSD for suppressing appearance-sensitive shallow features, and (3) SAAM for stability estimation and selective alignment. SAAM computes the stability score $d_{\text{stab}}(i)$, selects the Top- k stable regions to form $W(i)$, and combines it with a foreground prior to obtain the final weight $A(i)$. The projected deep features are then selectively aligned under the guidance of the final weight $A(i)$.

than explicitly modeling whether a region remains semantically stable across heterogeneous perturbations before alignment. Second, coarse spatial priors cannot capture local spatial variation in regional stability [17, 53]. These gaps motivate stability-aware alignment for consistency learning.

3 Proposed Method

Problem formulation. We study single-source domain generalization (SSDG) for medical image segmentation. Let $\mathcal{D}_s = \{(x_i, y_i)\}_{i=1}^N$ denote a labeled source dataset with feature space $\mathcal{X}$ and label space $\mathcal{Y}$. The goal is to learn a segmenter $f_\theta : \mathcal{X} \rightarrow \mathcal{Y}$ that generalizes to unseen target domains without accessing target domain data during training.

Overview. As illustrated in Fig. 2, SAA consists of three core components: Tri-View Construction (Sec. 3.1), Channel-Gated Structure-Style Decoupling (CGSD, Sec. 3.2), and the Stability-Aware Alignment Module (SAAM, Sec. 3.3). Tri-View Construction generates an anchor view and two perturbed views with progressively stronger appearance variation. CGSD suppresses shallow appearance-sensitive channels to improve the reliability of downstream stability estimation. Based on the resulting deep features, SAAM performs stability estimation, stable region selection, and distance-weighted alignment over selected stable locations. Together, these components establish SAA as a stability-aware representation learning framework for single-source domain generalization.

3.1 Tri-View Construction

To explicitly model semantic stability under controlled perturbations, for each input image $x \in \mathbb{R}^{H \times W}$, we construct three views

$x^{(k)} = \mathcal{T}_k(x)$, $k \in \{0, 1, 2\}$, corresponding to the anchor, base, and strong views. The anchor view is unchanged, the base view introduces mild global intensity perturbation, and the strong view applies stronger cascaded appearance perturbations.

Anchor View. The anchor view is defined as $x^{(0)} = x$ and serves as the semantic reference.

Base View (GIP). The base view introduces global intensity perturbation (GIP) to simulate moderate cross-domain appearance shifts:

$$x^{(1)} = \alpha F(x) + \beta, \quad (1)$$

where $\alpha \sim \mathcal{TN}(1, \sigma_1)$ and $\beta \sim \mathcal{TN}(0, \sigma_2)$ are scale and shift factors sampled from truncated Gaussian distributions to avoid excessively large intensity distortions. $F(\cdot)$ denotes a monotonic cubic Bézier intensity mapping that introduces non-linear intensity variations while preserving structural contrast relationships. The transformation is applied in the normalized intensity domain.

Strong View. To construct a stronger yet semantic-preserving view, we apply a cascaded perturbation pipeline consisting of class-conditional local perturbation (CLP), saliency-guided fusion (SGF), and random convolution candidate selection (RCCS).

Appearance variations across domains are typically spatially non-uniform, suggesting that different semantic regions may respond differently to perturbations. Accordingly, we first apply a class-conditional local perturbation (CLP) on the input x , as shown in Eq. (2), where each region is independently modulated via class-wise scaling and shifting.

$$x^{\text{clp}} = \sum_{c=1}^C (\alpha_c x_c + \beta_c m_c) \quad (2)$$

where $c \in \{1, \dots, C\}$ indexes the semantic class, m_c denotes the binary mask of class c derived from the ground-truth label y , $x_c = m_c \odot x$ represents the region of class c , and (α_c, β_c) are class-wise intensity scaling and shifting parameters, $\odot$ denotes element-wise multiplication.

To reduce excessive distortion in salient regions, we fuse x^{clp} with the base view $x^{(1)}$ through a saliency-guided fusion (SGF) mechanism:

$$\tilde{x}^{(2)} = S \odot x^{(1)} + (1 - S) \odot x^{\text{clp}}, \quad (3)$$

where $S = \mathcal{G}_{\text{sal}}(x^{(1)}) \in [0, 1]^{H \times W}$ denotes the saliency map computed from the gradient magnitude of the segmentation network with respect to the base-view input $x^{(1)}$. High-saliency locations receive more contribution from the base view $x^{(1)}$, whereas low-saliency locations receive more contribution from the class-conditionally perturbed result x^{clp} .

To further enrich local texture diversity, random convolution candidate selection (RCCS) is introduced. We generate K random-convolution candidates $\hat{x}_k = \mathcal{R}_k(\tilde{x}^{(2)})$ and select the one with the smallest cosine distance to the anchor representation:

$$x^{(2)} = \arg \min_{\hat{x} \in \{\hat{x}_1, \dots, \hat{x}_K\}} \mathcal{D}(f_{\theta}(\hat{x}), f_{\theta}(x^{(0)})). \quad (4)$$

where $f_{\theta}(\cdot)$ denotes the encoder feature extractor.

3.2 CGSD: Channel-Gated Structure-Style Decoupling

In medical image segmentation, shallow features often contain both relatively stable structural patterns and more variable appearance-related responses. However, as shown in Fig. 3, these properties are not uniformly distributed across channels: channels with low discrepancy tend to produce more consistent and spatially concentrated responses across views, whereas high-discrepancy channels exhibit more dispersed and variation-sensitive patterns. To suppress appearance-sensitive channels, we introduce a Channel-Gated Structure-Style Decoupling (CGSD) module with a learnable view-shared gating vector. CGSD emphasizes structure-dominant channels and reduces the contribution of style-sensitive ones before downstream encoding.

Given the cascaded views $\{x^{(0)}, x^{(1)}, x^{(2)}\}$, a shared convolutional stem extracts shallow features for each view: $f^{(k)} = f_{\theta}(x^{(k)}) \in \mathbb{R}^{B \times C \times H \times W}$, ($k = 0, 1, 2$). CGSD then introduces a learnable channel gate $\mathbf{g} \in \mathbb{R}^C$ shared across views. The structure channel weights are computed via a sigmoid function:

$$\mathbf{w}^{\text{str}} = \sigma(\mathbf{g}), \quad \mathbf{w}^{\text{str}} \in (0, 1)^C, \quad (5)$$

while style channel weights are complementary:

$$\mathbf{w}^{\text{sty}} = 1 - \mathbf{w}^{\text{str}}. \quad (6)$$

The feature map is decomposed as:

$$f_{\text{str}}^{(k)} = f^{(k)} \odot \mathbf{w}^{\text{str}}, \quad f_{\text{sty}}^{(k)} = f^{(k)} \odot \mathbf{w}^{\text{sty}}, \quad (7)$$

where $\odot$ denotes channel-wise multiplication with spatial broadcasting. Only $f_{\text{str}}^{(k)}$ is forwarded to subsequent encoder layers, while $f_{\text{sty}}^{(k)}$ is suppressed.

To ensure that the gating mechanism captures consistent structural cues across perturbations, we introduce a regularization between the base and strong views. A lightweight projector $\phi(\cdot)$ maps

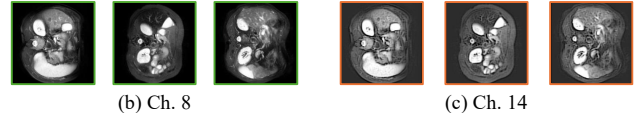
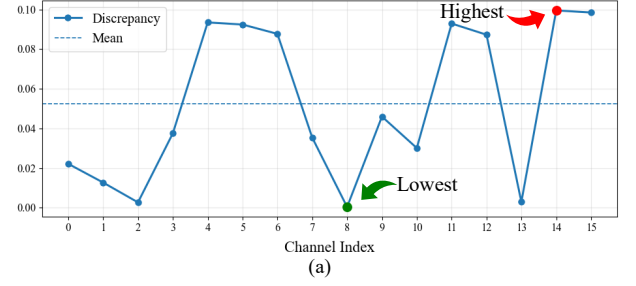


Figure 3: Channel-wise discrepancy analysis for CGSD. (a) Cross-view discrepancy scores of individual shallow channels. (b) Feature maps from the lowest-discrepancy channel. (c) Feature maps of the highest-discrepancy channel.

both $f_{\text{str}}^{(k)}$ and $f_{\text{sty}}^{(k)}$ into a low-dimensional embedding space. We impose two complementary constraints: (1) Structure consistency, which encourages the embeddings of $f_{\text{str}}^{(k)}$ to remain similar across perturbations; and (2) Style diversity, which encourages the embeddings of $f_{\text{sty}}^{(k)}$ to remain distinguishable, reflecting appearance variation. The corresponding loss is formulated as:

$$\mathcal{L}_{\text{cgsd}} = d(\phi(f_{\text{str}}^{(1)}), \phi(f_{\text{str}}^{(2)})) - d(\phi(f_{\text{sty}}^{(1)}), \phi(f_{\text{sty}}^{(2)})). \quad (8)$$

Here, $d(\cdot, \cdot)$ denotes cosine distance.

By enforcing channel-wise allocation guided by cross-view consistency, CGSD produces more reliable feature representations for subsequent modules.

3.3 Stability-Aware Alignment Module (SAAM)

While CGSD emphasizes stable representations via channel-wise allocation, cross-view consistency also depends on spatial alignment. Spatially unstable responses may still persist at certain locations, and enforcing consistency everywhere may introduce unreliable signals. SAAM therefore estimates spatial stability across views and restricts alignment to reliable regions.

Given deep encoder features $\{F^{(k)}\}_{k=0}^2$, where $F^{(k)} \in \mathbb{R}^{C_f \times H_f \times W_f}$, SAAM performs three steps: stability estimation, stable-region selection, and anchor-centered weighted alignment. Stability is estimated on the deep feature maps at resolution $H_f \times W_f$. The base-strong term is used only for stability estimation, while the final alignment remains anchor-centered.

Let $F_i^{(k)} \in \mathbb{R}^{C_f}$ denote the feature vector of view k at spatial location i . We measure cross-view stability using pairwise cosine dissimilarities:

$$d_{kl}(i) = 1 - \frac{\langle F_i^{(k)}, F_i^{(l)} \rangle}{\|F_i^{(k)}\|_2 \|F_i^{(l)}\|_2}, \quad (k, l) \in \{(0, 1), (0, 2), (1, 2)\}, \quad (9)$$

and define the stability score as

$$d_{\text{stab}}(i) = \frac{1}{3} \sum_{(k,l)} d_{kl}(i), \quad (10)$$

where smaller $d_{\text{stab}}(i)$ indicates higher cross-view stability. We select the Top- ρ most stable locations by

$$\Omega(i) = \mathbb{I}(i \in \text{TopK}_\rho(-d_{\text{stab}})), \quad (11)$$

where $\mathbb{I}(\cdot)$ is the indicator function and ρ is the selection ratio. We further assign a soft reliability weight

$$R(i) = \exp\left(-\frac{d_{\text{stab}}(i)}{\tau}\right), \quad W(i) = \Omega(i) \cdot R(i). \quad (12)$$

The resulting $W(i) \in [0, 1]$ serves as a low-resolution stability gate for alignment.

To suppress evidently irrelevant background positions, we further combine W with a foreground union prior M :

$$M(i) = \mathbb{I}[(y_i \neq 0) \vee (\hat{y}_i^{(0)} \neq 0) \vee (\hat{y}_i^{(1)} \neq 0) \vee (\hat{y}_i^{(2)} \neq 0)], \quad (13)$$

where y is the ground-truth label map and $\hat{y}^{(k)}$ denotes the prediction of view k . The final alignment weight is

$$A(i) = \text{Up}_{\text{bi}}(W(i)) \cdot M(i), \quad (14)$$

where $\text{Up}_{\text{bi}}(\cdot)$ denotes bilinear upsampling and M acts as a conservative spatial filter during optimization.

For alignment, we first map the deep features into a compact embedding space:

$$q^{(k)} = g_\phi(F^{(k)}), \quad q^{(k)} \in \mathbb{R}^{C_q \times H_q \times W_q}. \quad (15)$$

The projected features are then upsampled to the alignment resolution:

$$\hat{q}^{(k)} = \text{Up}_{\text{bi}}(q^{(k)}), \quad \hat{q}^{(k)} \in \mathbb{R}^{C_q \times H \times W}. \quad (16)$$

Using the final pixel-wise weight A , we enforce anchor-centered alignment on the projected features by minimizing the anchor-to-base and anchor-to-strong cosine distances:

$$\mathcal{L}_{\text{saam}} = \frac{\sum_i A(i) \cdot (d(\hat{q}_i^{(0)}, \hat{q}_i^{(1)}) + d(\hat{q}_i^{(0)}, \hat{q}_i^{(2)}))}{\sum_i A(i) + \epsilon}, \quad (17)$$

where $d(\cdot, \cdot)$ denotes cosine distance and ϵ is a small constant for numerical stability.

4 Experiments

4.1 Experimental Settings

4.1.1 Datasets. We evaluate SAA on two public medical image SSDG benchmarks, covering abdominal cross-modality transfer and cardiac cross-sequence transfer, under the strict single-source setting. The abdominal benchmark is constructed from BTCV CT and CHAOS T2-weighted MRI data [13, 16], while the cardiac benchmark is based on the MS-CMRSeg dataset [52]. For each benchmark, one modality or sequence is used as the only source domain for training, and the other is used only for testing.

4.1.2 Implementation Details. We adopt a standard U-Net as the segmentation backbone in all experiments, the same as CGSD [30]. The segmentation model is trained with the Adam optimizer. All models are trained from scratch with an initial learning rate of 5×10^{-4} linearly decayed to zero and a batch size of 20. We train for 1500 epochs on the abdominal benchmark and 1800 epochs on the cardiac benchmark on a single NVIDIA GeForce RTX 3090 GPU with 24 GB memory.

4.1.3 Comparison Methods. We compare SAA with representative baselines under the strict single-source setting, including generic augmentation-based DG methods (Cutout [6], MixStyle [50], AdvBias [4], and RandConv[43]), a feature-regularization baseline (RSC [11]), and recent SSDG methods (CSDG [30], SLAug [35], S2S2 [31], and DCON [38]), together with the ERM baseline. All methods use the same backbone, data split, and optimization protocol for fair comparison.

4.2 Experimental Results

Table 1 reports the Dice scores on all four SSDG tasks. SAA achieves the best average Dice on all tasks, outperforming ERM, generic DG baselines, and recent SSDG methods, which indicates stronger robustness under both cross-modality and cross-sequence shifts.

4.2.1 Comparisons on Abdominal (BTCV-CHAOS). On the abdominal benchmark, SAA achieves the best average Dice on both MRI→CT and CT→MRI. Compared with ERM, SAA improves the average Dice by 22.57 points on MRI→CT (85.55 vs. 62.98) and by 11.94 points on CT→MRI (89.20 vs. 77.26). Compared with the strongest augmentation baseline RandConv, the gains are 8.99 and 8.54 points, respectively; compared with the feature-regularization baseline RSC, the gains are 19.48 and 15.11 points. Against recent SSDG methods, SAA further surpasses the best competitor by 1.10 points on MRI→CT (vs. DCON) and 0.57 points on CT→MRI (vs. SLAug). Notably, SAA achieves the best scores for all abdominal organs in both transfer directions, indicating consistent improvements rather than gains concentrated on a single structure. The advantage is particularly clear on the harder MRI→CT task, where large appearance discrepancies make purely augmentation-based coverage insufficient. These results show that simply increasing appearance diversity or applying global feature regularization is insufficient under severe cross-modality shifts. By explicitly estimating cross-view stability and restricting alignment to reliable regions, SAA produces more robust anatomical representations.

4.2.2 Comparisons on Cardiac (MS-CMRSeg). On the cardiac benchmark, SAA also achieves the best average Dice on both LGE→bSSFP and bSSFP→LGE. Compared with ERM, SAA improves the average Dice by 2.64 points on LGE→bSSFP (88.12 vs. 85.48) and by 11.43 points on bSSFP→LGE (87.33 vs. 75.90). Compared with RandConv, the gains are 0.88 and 3.60 points, and compared with RSC, the gains are 2.52 and 9.82 points, respectively. SAA further improves over the strongest recent SSDG baseline by 0.06 points on LGE→bSSFP (vs. DCON) and 0.64 points on bSSFP→LGE (vs. SLAug), while also achieving the best MYO and RVC scores on bSSFP→LGE. On LGE→bSSFP, the margin in average Dice is smaller, but SAA still ranks first overall, showing that the proposed selective alignment remains beneficial even when competing methods are already strong. On bSSFP→LGE, the larger gain suggests that SAA is more effective when the target-domain appearance is more challenging and local responses become less reliable. These results suggest that SAA is especially beneficial when local appearance variation is strong and anatomical boundaries are ambiguous, since stability-aware selective alignment helps suppress unreliable responses while preserving shape-consistent cues.

Table 1: Comparison with different methods. Dice (%) is used as the evaluation metric. Best in bold; second-best underlined.

Method	Abdominal MRI-CT					Cardiac LGE-bSSFP			
	Liver	R-Kidney	L-Kidney	Spleen	Average	LVC	MYO	RVC	Average
Supervised	98.90	92.38	92.21	89.08	93.14	91.47	83.20	90.59	88.42
ERM	88.28	42.74	64.33	56.57	62.98	89.73	78.66	88.06	85.48
Cutout	86.99	63.66	73.74	57.60	70.50	90.88	79.14	87.74	85.92
RSC	88.10	46.60	75.94	53.61	66.07	90.21	78.63	87.96	85.60
MixStyle	86.66	48.26	65.20	55.68	63.95	91.22	79.64	88.16	86.34
AdvBias	87.63	52.48	68.28	50.95	64.84	91.20	79.50	88.10	86.27
RandConv	84.14	76.81	77.99	67.32	76.56	91.98	80.92	88.83	87.24
CSDG	85.62	80.02	80.42	75.56	80.40	91.37	80.43	89.16	86.99
SLAug	89.26	80.98	82.05	79.93	83.05	91.92	81.49	89.61	87.67
S2S2	88.30	81.79	80.31	82.20	83.15	92.16	80.19	90.10	87.49
DCON	<u>89.98</u>	<u>82.28</u>	<u>82.95</u>	<u>82.60</u>	<u>84.45</u>	92.53	<u>81.60</u>	90.05	<u>88.06</u>
SAA (ours)	90.20	83.10	83.40	85.50	85.55	<u>92.43</u>	81.86	<u>90.07</u>	88.12

Method	Abdominal CT-MRI					Cardiac bSSFP-LGE			
	Liver	R-Kidney	L-Kidney	Spleen	Average	LVC	MYO	RVC	Average
Supervised	91.63	92.59	89.95	90.22	91.10	91.96	83.61	90.05	88.54
ERM	77.66	78.40	78.21	74.75	77.26	85.58	67.33	74.80	75.90
Cutout	79.80	82.32	82.14	76.24	80.12	88.35	69.06	79.19	78.87
RSC	76.40	75.79	76.60	67.56	74.09	87.06	69.77	75.69	77.51
MixStyle	77.63	78.41	78.03	77.12	77.80	85.78	64.23	75.61	75.21
AdvBias	78.54	81.70	80.69	79.73	80.17	88.23	70.29	80.32	79.62
RandConv	73.63	79.69	85.89	83.43	80.66	89.88	75.60	85.70	83.73
CSDG	86.62	87.48	86.88	84.27	86.31	90.35	77.82	86.87	85.01
SLAug	90.08	<u>89.23</u>	<u>87.54</u>	<u>87.67</u>	<u>88.63</u>	91.53	<u>80.65</u>	<u>87.90</u>	<u>86.69</u>
S2S2	<u>90.71</u>	89.21	86.55	84.50	87.75	<u>91.48</u>	79.84	86.87	86.06
DCON	88.75	87.15	85.89	86.80	87.15	90.80	78.40	86.82	85.34
SAA (ours)	90.95	89.79	88.10	87.95	89.20	91.44	81.87	88.67	87.33

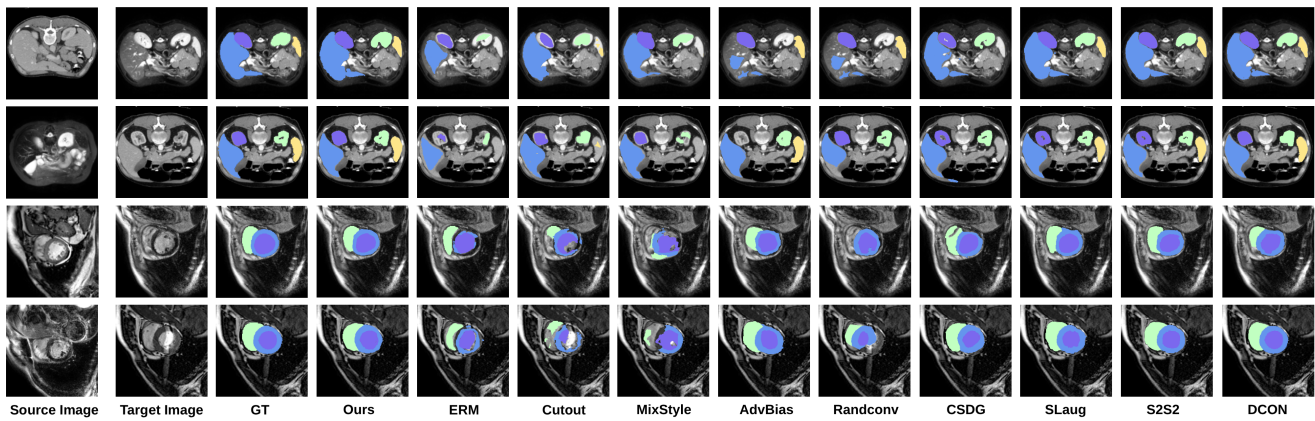
**Figure 4: Qualitative segmentation comparison on unseen target domains. Top two rows show Abdominal MRI-CT and CT-MRI, while bottom two rows show Cardiac bSSFP-LGE and LGE-bSSFP.**

Figure 4 provides qualitative comparisons on unseen target domains. We select four representative target-domain cases as examples. Compared with representative baselines, SAA yields cleaner

anatomical boundaries and fewer false positives on unseen target

Table 2: Ablation results (Dice %) on Abdominal MRI→CT and Cardiac LGE→bSSFP. Single-module variants use Baseline + one module, while Full uses all modules.

Method	Abd. MRI→CT					LGE→bSSFP			
	Liv.	R-Kid.	L-Kid.	Spl.	Avg.	LVC	MYO	RVC	Avg.
Baseline	87.89	81.55	80.31	81.33	82.77	90.74	80.49	88.39	86.54
+RCCS	88.34	82.14	81.59	82.12	83.55	91.11	81.07	89.63	87.27
+CGSD	88.32	82.01	81.06	81.73	83.28	91.46	81.10	88.94	87.17
+SAAM	89.35	82.17	82.34	84.89	84.69	92.00	81.78	89.39	87.72
Full	90.20	83.10	83.40	85.50	85.55	92.43	81.86	90.07	88.12

Table 3: Ablation results (Dice %) on Abdominal CT→MRI and Cardiac bSSFP→LGE.

Method	Abd. CT→MRI					bSSFP→LGE			
	Liv.	R-Kid.	L-Kid.	Spl.	Avg.	LVC	MYO	RVC	Avg.
Baseline	89.08	88.72	86.13	85.04	87.24	90.41	80.19	86.04	85.55
+RCCS	89.36	88.96	86.51	85.96	87.70	90.81	80.67	87.07	86.18
+CGSD	89.40	89.04	86.87	85.76	87.77	90.94	80.37	87.23	86.18
+SAAM	90.35	89.40	87.10	86.78	88.41	91.27	81.31	87.71	86.76
Full	90.95	89.79	88.10	87.95	89.20	91.44	81.87	88.67	87.33

domains, especially in low-contrast regions and around ambiguous structures. These visual results are consistent with the quantitative improvements and further demonstrate the superior generalization capability of SAA.

4.3 Ablation and Analysis

We further analyze SAA through component ablation 4.3.1, SAAM core ablation 4.3.2, tri-view validation 4.3.3, hyper-parameter sensitivity 4.3.4, and mechanism verification 4.3.5 under the same training protocol as the full model.

4.3.1 Component Ablation. Tables 2 and 3 report the component ablation results on all four SSDG tasks. All three modules improve over the baseline, with SAAM consistently giving the largest gain among the single-module variants, while the Full model achieves the best performance in every task.

Specifically, the Full model improves the average Dice from 82.77 to 85.55 on MRI→CT (+2.78), from 86.54 to 88.12 on LGE→bSSFP (+1.58), from 87.24 to 89.20 on CT→MRI (+1.96), and from 85.55 to 87.33 on bSSFP→LGE (+1.78). In comparison, SAAM alone already yields strong gains of +1.92, +1.18, +1.17, and +1.21 on the four tasks, respectively. These results indicate that SAAM is the strongest individual contributor, while RCCS and CGSD provide further complementary improvements in the full model.

4.3.2 SAAM Core Ablation. To isolate the effect of each SAAM component, we remove the alignment term, the stable-region selector Ω , or the distance-aware weighting W while keeping the remaining parts unchanged. As shown in Tables 4 and 5, all three variants underperform the full model on both task-level and organ-level results. At the task level, removing the alignment term reduces the overall Dice from 87.55 to 86.30, while removing Ω and W leads to smaller drops to 86.70 and 86.98, respectively. The same trend is observed on MRI→CT and LGE→bSSFP, where the full model achieves the best average Dice of 85.55 and 88.12. Removing the alignment term causes the largest drop, while removing Ω is more

Table 4: SAAM core ablation. Removing Ω , W , or the alignment term reduces average Dice.

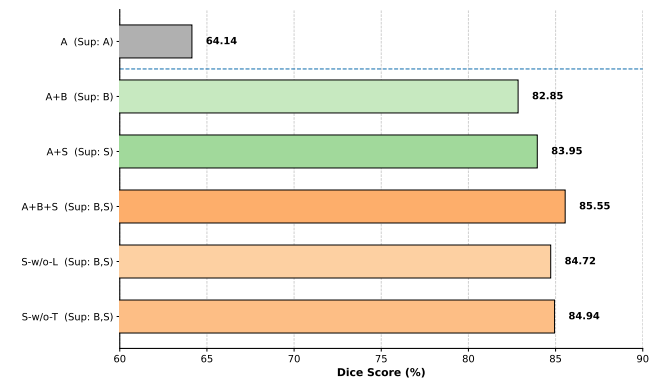
Method	MRI→CT	CT→MRI	LGE→bSSFP	bSSFP→LGE	Overall
Full	85.55	89.20	88.12	87.33	87.55
w/o Ω	84.85	88.35	87.55	86.05	86.70
w/o W	85.20	88.60	87.78	86.35	86.98
w/o align	84.35	87.90	87.20	85.75	86.30

Table 5: Organ-level SAAM core ablation on MRI→CT and LGE→bSSFP.

Method	MRI→CT					LGE→bSSFP			
	Liver	R-Kid.	L-Kid.	Spleen	Avg.	LVC	MYO	RVC	Avg.
Full	90.20	83.10	83.40	85.50	85.55	92.43	81.86	90.07	88.12
w/o Ω	89.90	82.30	82.70	84.50	84.85	92.10	81.00	89.55	87.55
w/o W	90.05	82.65	82.95	85.15	85.20	92.19	81.28	89.87	87.78
w/o align	89.70	81.85	82.10	83.75	84.35	92.00	80.70	88.90	87.20

harmful than removing W , indicating that stable-region selection plays a more important role than distance modulation in suppressing unstable responses before alignment.

4.3.3 Effectiveness of Tri-view. Fig. 5 evaluates the tri-view design with CGSD and SAAM fixed. The full configuration (A+B+S) achieves the best performance, while both A+B and A+S outperform the anchor-only setting. Removing either its local component (S-w/o-L) or texture component (S-w/o-T) degrades performance, showing that effective strong perturbation requires joint modeling of local semantics and texture. These results show that the perturbation hierarchy (A→B→S) provides more reliable supervision for stability-aware alignment than isolated augmentations.

**Figure 5: Ablation of tri-view construction on MRI→CT (Dice, %). A: anchor; B: base; S: strong view. All variants share the same alignment mechanism.**

4.3.4 Hyper-parameter Analysis. Fig. 7 shows the sensitivity of the Top- k ratio in stable-region selection. Performance peaks at Top- $k = 0.3$, which we use in all experiments. Smaller ratios retain too few stable locations for effective alignment, while larger ratios gradually include less reliable boundary or ambiguous regions.

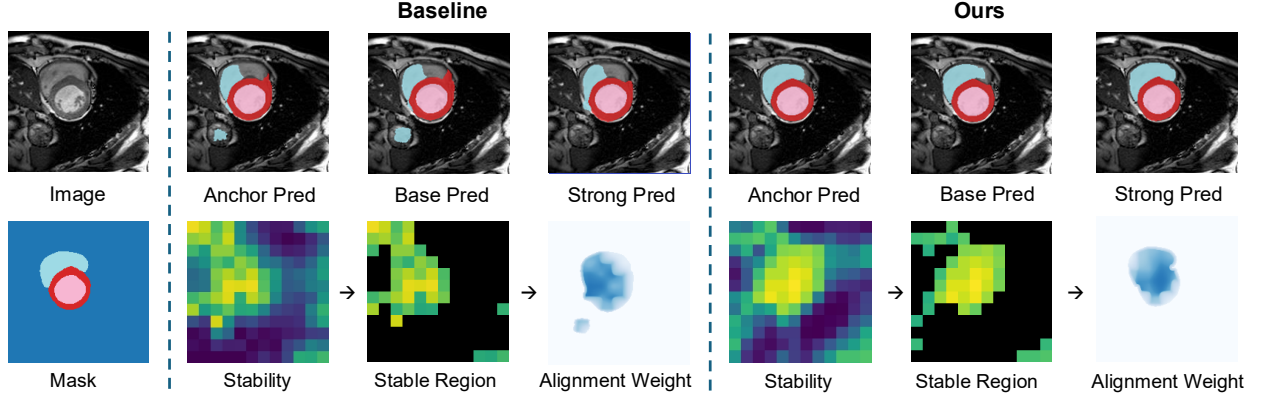


Figure 6: Mechanism verification of stability-guided alignment. For visual comparability, the same input image is used as the background, while the overlaid contours correspond to predictions from the anchor, base, and strong views. We compare the baseline model without SAAM and the full SAA model. The bottom row visualizes the stability response W , the selected stable region Ω , and the final alignment weight A . Compared with the baseline, SAA yields more consistent cross-view predictions. In the bottom row, the baseline shows less organ-centered stability distributions, whereas our method produces a more organ-centered stability pattern and compact stable cores. After being combined with foreground constraints, these cores are transformed into smoother and more anatomically focused alignment weights. This progressive refinement explains how SAA improves consistency while avoiding over-alignment on unstable regions.

The relatively flat curves around the optimum indicate moderate robustness to the choice of Top- k . This trend is consistent with Fig. 6, where effective alignment depends on preserving compact stable cores while excluding noisy peripheral responses.

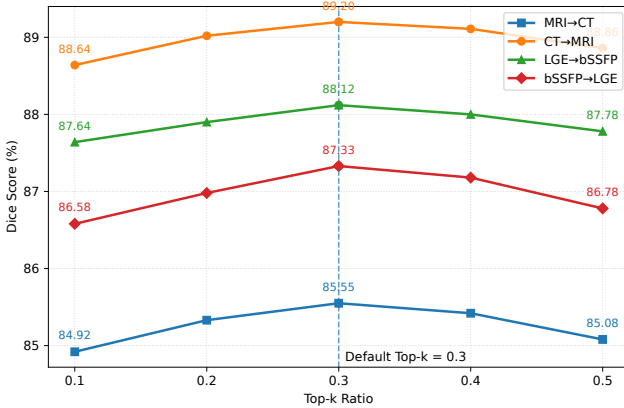


Figure 7: Sensitivity analysis of the Top- k ratio in Stable Region Selection.

4.3.5 Mechanism Verification: Stability-guided alignment. To analyze how SAA affects cross-view consistency learning, we visualize the predictions from different views, the stability map W , the selected stable region Ω , and the final alignment weight A under anchor, base, and strong perturbations. As shown in Fig. 6, the baseline exhibits larger inter-view prediction variation, while our method produces more consistent predictions across perturbed views. This difference is reflected not only in the contour overlap

across views, but also in the spatial concentration of W , Ω , and A toward structurally reliable organ interiors. More importantly, the advantage comes from how SAAM converts estimated stability into compact stable-core selection and foreground-constrained alignment weights. Instead of enforcing equally strong consistency over all foreground responses, SAAM concentrates alignment on more reliable organ-centered regions. It suppresses overly aggressive alignment on boundary-blurred, locally conflicting, or perturbation-sensitive locations.

5 Conclusion

In this work, we revisit the fundamental assumption underlying single-source domain generalization for medical image segmentation. While most existing approaches implicitly rely on strong augmentations and enforce invariance across all perturbed views, our findings highlight a critical limitation of this paradigm. Medical image augmentations are not always safe. Due to the low contrast, subtle inter-class differences, and anatomical sensitivity inherent to medical images, certain perturbations may selectively distort semantically critical regions. Treating all perturbed views as equally reliable and enforcing unconditional invariance can therefore lead to erroneous alignment and compromised generalization. Instead, we demonstrate that effective generalization under single-source constraints depends on identifying which semantic information remains stable across different views and aligning only these reliable components. By shifting the focus from unconditional invariance to stability-aware alignment, our work offers a principled perspective on how transferable representations should be learned for medical image segmentation. We hope this perspective encourages future research to reconsider the role of data augmentation and consistency constraints in medical image analysis, and to move toward reliability-aware and stability-aware generalization strategies.

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